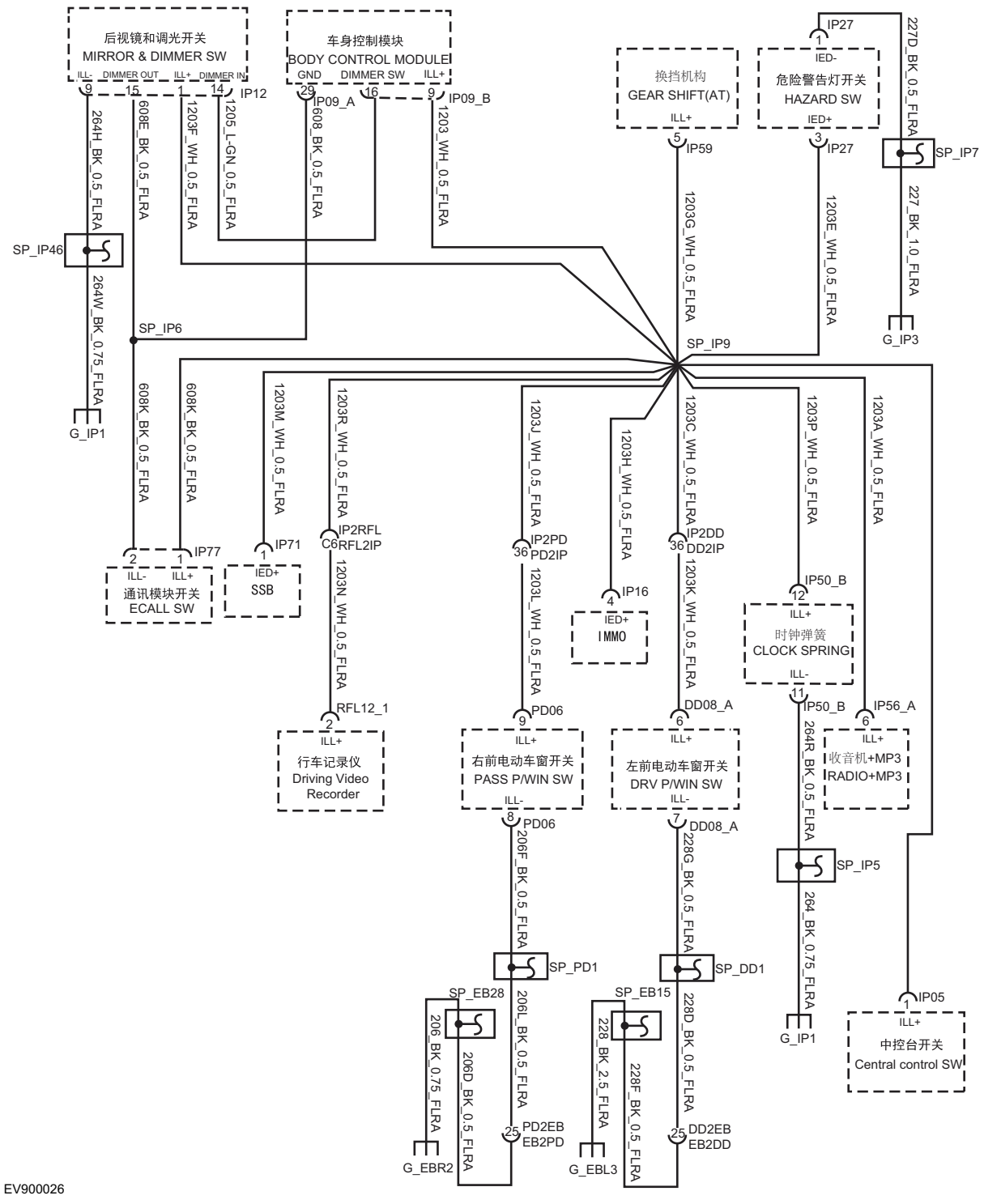
B1120-11、 B1120-12、 B1120-13

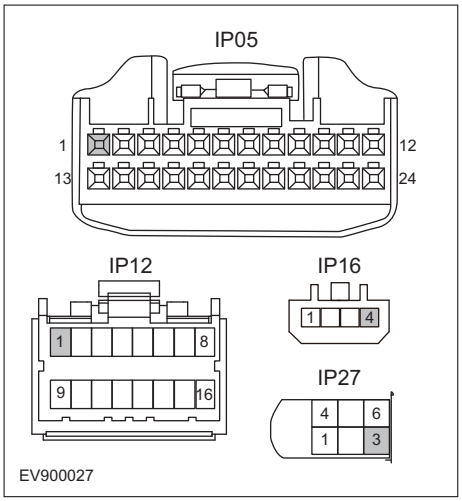
Fault diagnosis code
B1120-11：Backlight circuit short to ground
B1120-12：Backlight circuit short to battery
B1120-13：Backlight circuit open
Detection tools
1 Multimeter and diagnostic apparatus
Possible causes
• Circuit,terminal and adapting piece fault • BCM fault • Insurance fuse fault
Conditions for fault code operation
Turn the ignition switch to ON position.
Conditions for fault code setting
B1120-11
Back light circuit short to ground.
B1120-12
Back light circuit short to battery.
B1120-13
Back light circuit open
To set the effect of a fault code condition
Back light function fails
Description of circuit diagram

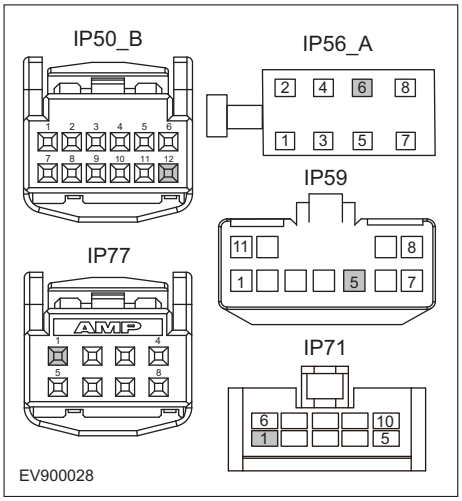
Circuit diagram

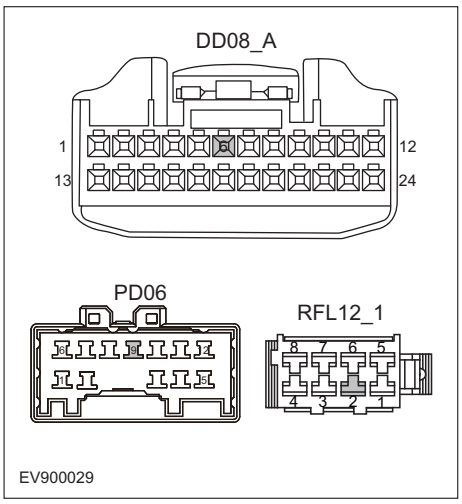


Troubleshooting steps		
Step 1	General inspection.	
		1 Check the fuse for damage. 2 Check battery capacity. 3 Check the harness plug of air conditioning control module for damage, poor contact, aging and looseness. 4 Check whether the result is normal or not?
		NO → Repair or replace the faulted parts.
YES ↓		
Step 2	Check fault codes.	
		1 Connect the fault diagnosis instrument to the diagnosis interface. 2 Turn the ignition switch to ON position. 3 Check the body control module with a diagnostic apparatus. 4 Is there any other fault code except for B1120-11/B1120-12/B1120-13?
		YES → Refer to:DTC Summary list(BCM).
NO ↓		

Step 3	Check the power voltage of back light when it is started or operating.







1 Turn the ignition switch to OFF.

2 Disconnect harness plug IP05 of center console switch, IP12 of rearview mirror and dimmer switch, IP16 of IMMO, IP27 of danger warning light switch, IP50_B of timepiece spring, IP56_A of radio and MP3 module, IP59 of shifting mechanism, IP71 of SSB, IP77 of communication module switch, RFL12_1 of automatic data recorder, PD06 of right front electric window, DD08_A of left front electric window.

3 Turn the ignition switch to ON.

4 Measure the voltage between the harness terminal of background light and ground.

Measuring circuit: voltage between terminal 1 on plug IP05 and ground.

Measuring circuit: voltage between terminal 1 on plug IP12 and ground.

Measuring circuit: voltage between terminal 4 on plug IP16 and ground.

Measuring circuit: voltage between terminal 3 on plug IP27 and ground.

Measuring circuit: voltage between terminal 12 on plug IP50_B and ground.

Measuring circuit: voltage between terminal 6 on plug IP56_A and ground.

Measuring circuit: voltage between terminal 5 on plug IP59 and ground.

Measuring circuit: voltage between terminal 1 on plug IP71 and ground.

Measuring circuit: voltage between terminal 1 on plug IP77 and ground.

Measuring circuit: voltage between terminal 2 on plug RFL12_1 and ground.

Measuring circuit: voltage between terminal 9 on plug PD06 and ground.

Measuring circuit: voltage between terminal 6 on plug DD08_A and ground.

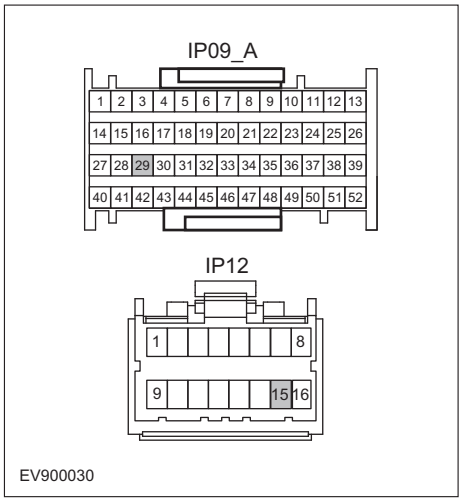
Standard value:10~14V

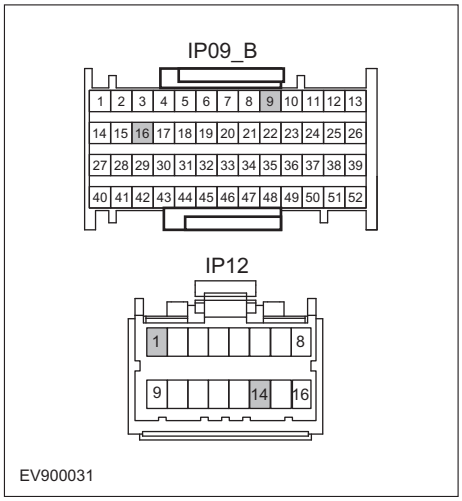
5 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 4	Check the circuit between rearview mirror and dimmer switch and BCM (open circuit).





1 Turn the ignition switch to OFF.

2 Disconnect the harness plug IP09_A, IP09_B of BCM module.

3 Measure the resistance between the harness terminal of step lamp and the BCM.

Measuring circuit: resistance between terminal 15 on plug IP12 and terminal 29 on plug IP09_A.

Measuring circuit: resistance between terminal 1 on plug IP12 and terminal 9 on plug IP09_B.

Measuring circuit: resistance between terminal 14 on plug IP12 and terminal 16 on plug IP09_B.

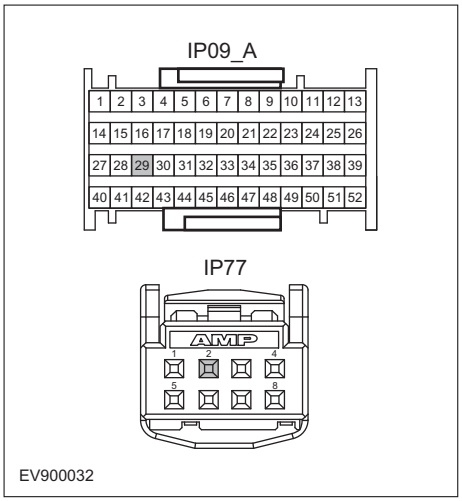
Standard value: ≈ 0Ω

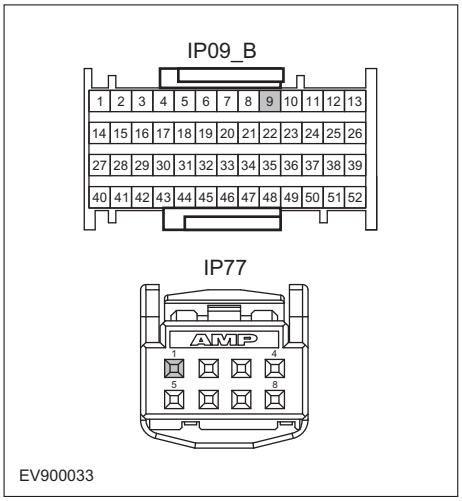
4 Check whether the result is normal or not?

NO → Repair or replace the faulted parts.

YES ↓

Step 5	Check the circuit between communication module switch and BCM (open circuit).





1 Measure the resistance between the harness terminal of communication module switch and the BCM.

Measuring circuit: resistance between terminal 2 on plug IP77 and terminal 29 on plug IP09_A.

Measuring circuit: resistance between terminal 1 on plug IP77 and terminal 9 on plug IP09_B.

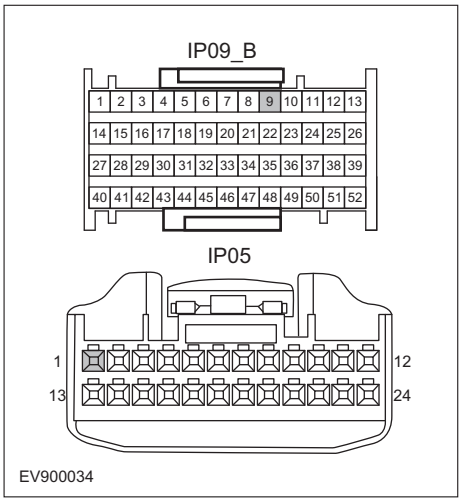
Standard value: ≈ 0Ω

2 Check whether the result is normal or not?

	NO → Repair or replace the faulted parts.

YES ↓

Step 6	Check the circuit between center console switch and BCM(open).



1 Measure the resistance between the harness terminal of center console switch and the BCM.

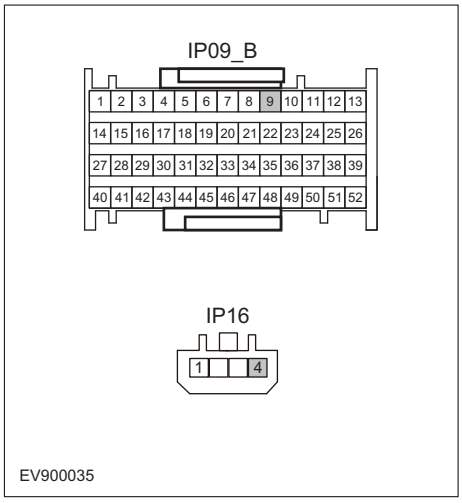
Measuring circuit: resistance between terminal 1 on plug IP05 and terminal 9 on plug IP09_B.

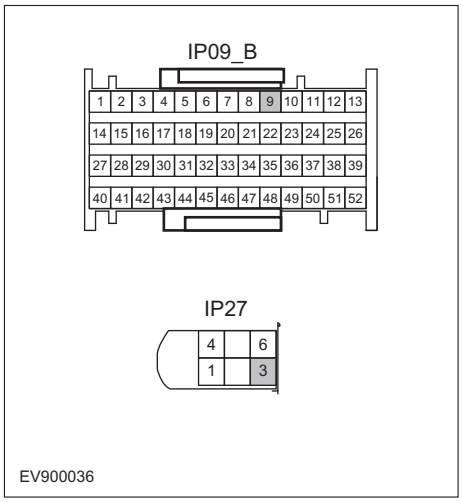
Standard value: ≈ 0Ω

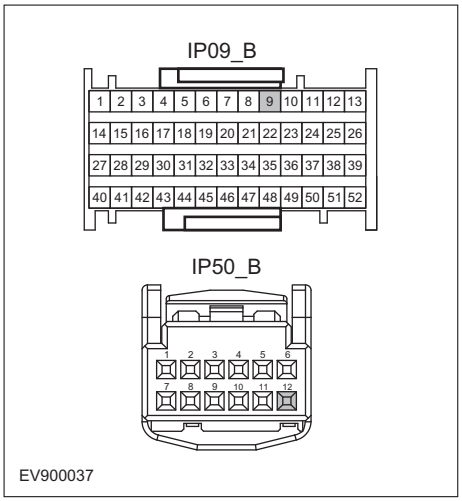
2 Check whether the result is normal or not?

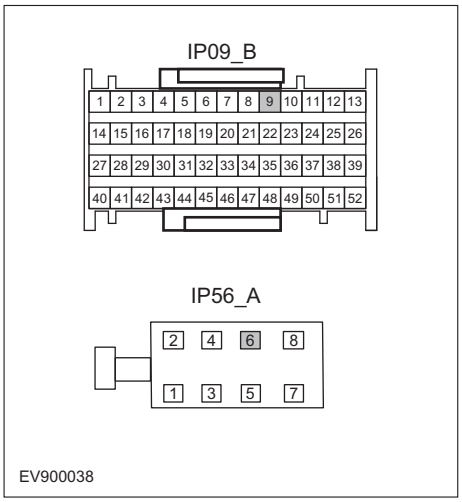
	NO → Repair or replace the faulted parts.

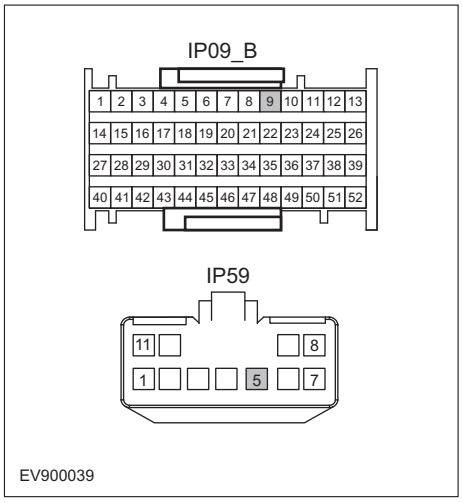
YES ↓

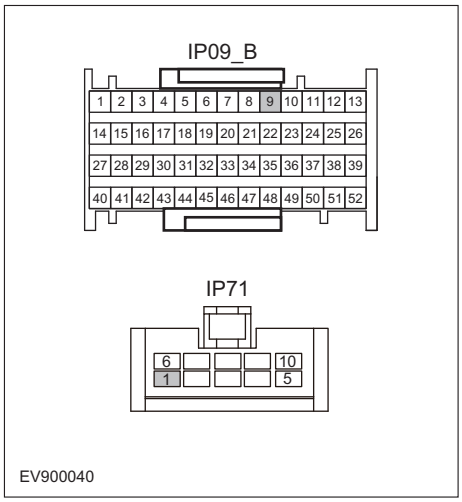
Step 7	Check the circuit between IMMO and BCM(open).
	1 Measure the resistance between the harness terminal of IMMO and the BCM. Measuring circuit: resistance between terminal 4 on plug IP16 and terminal 9 on plug IP09_B. **Standard value: ≈ 0Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

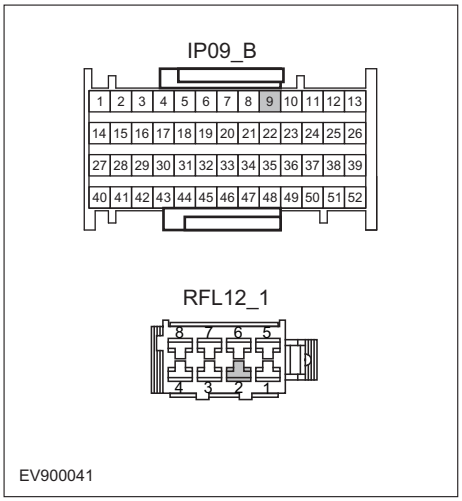
Step 8	Check the circuit between danger warning light switch and BCM(open).
	1 Measure the resistance between the harness terminal of danger warning light switch and the BCM. Measuring circuit: resistance between terminal 3 on plug IP27 and terminal 9 on plug IP09_B. **Standard value: ≈ 0Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

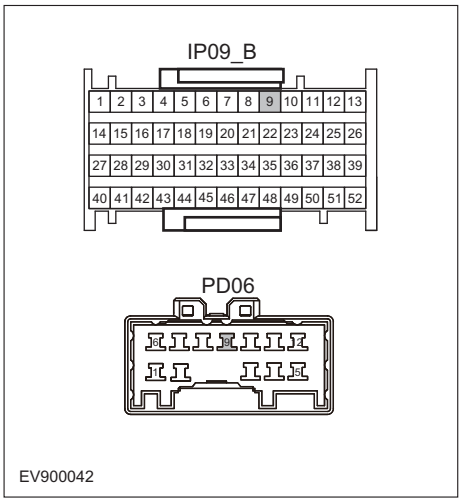
Step 9	Check the circuit between timepiece spring and BCM(open).
	1 Measure the resistance between the harness terminal of timepiece spring and the BCM. Measuring circuit: resistance between terminal 12 on plug IP50_B and terminal 9 on plug IP09_B. **Standard value: ≈ 0Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

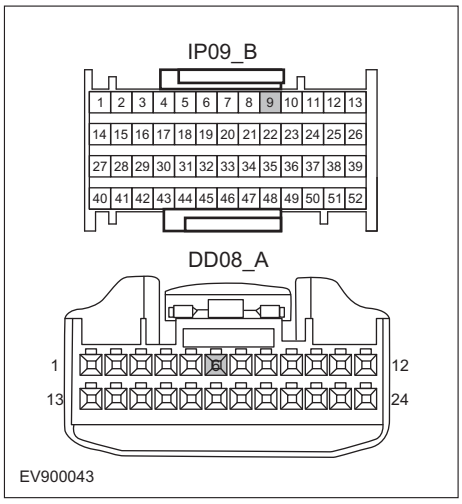
Step 10	Check the circuit between radio+ MP3 and BCM (open).
	1 Measure the resistance between the harness terminal of radio+ MP3 and of BCM. Measuring circuit: resistance between terminal 6 on plug IP56_A and terminal 9 on plug IP09_B. **Standard value: ≈ 0Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

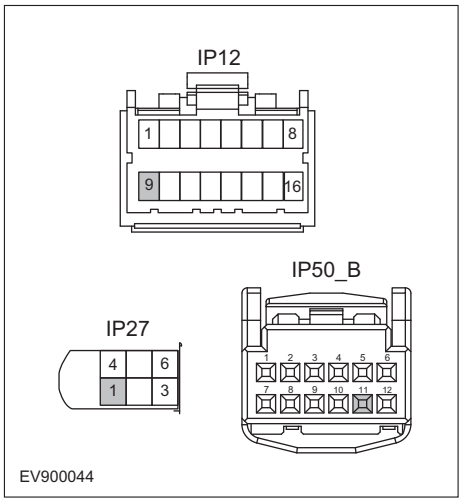
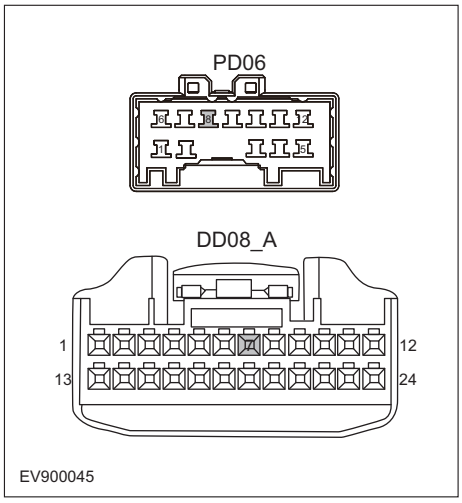
Step 11	Check the circuit between shifting mechanism and BCM(open).
	1 Measure the resistance between the harness terminal of shifting mechanism and the BCM. Measuring circuit: resistance between terminal 5 on plug IP59 and terminal 9 on plug IP09_B. **Standard value: ≈ 0Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 12	Check the circuit between SSB and BCM (open).
	1 Measure the resistance between the harness terminal of SSB and the BCM. Measuring circuit: resistance between terminal 1 on plug IP71 and terminal 9 on plug IP09_B. **Standard value: ≈ 0Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 13	Check the circuit between automatic data recorder and BCM(open).
	1 Measure the resistance between the harness terminal of automatic data recorder and the BCM. Measuring circuit: resistance between terminal 2 on plug RFL12_1 and terminal 9 on plug IP09_B. **Standard value: ≈ 0Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 14	Check the circuit between right front electric window switch and BCM(open).
	1 Measure the resistance between the harness terminal of right front electric window switch and the BCM. Measuring circuit: resistance between terminal 9 on plug PD06 and terminal 9 on plug IP09_B. **Standard value: ≈ 0Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 15	Check the circuit between left front electric window switch and BCM(open).
	1 Measure the resistance between the harness terminal of right front electric window switch and the BCM. Measuring circuit: resistance between terminal 6 on plug DD08_A and terminal 9 on plug IP09_B. **Standard value: ≈ 0Ω** 2 Check whether the result is normal or not?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 16	Test whether the grounding circuit of background light is open.
 	1 Turn the ignition switch to OFF. 2 Measure the resistance between the harness terminal of background light and ground. Measuring circuit: resistance between terminal 9 on plug IP12 and ground. Measuring circuit: resistance between terminal 1 on plug IP27 and ground. Measuring circuit: resistance between terminal 11 on plug IP50_B and ground. Measuring circuit: resistance between terminal 8 on plug PD06 and ground. Measuring circuit: resistance between terminal 7 on plug DD08_A and ground. **Standard value: < 1Ω** 3 Test whether the resistance is less than 1Ω?
	NO → Repair or replace the faulted parts.
YES ↓	

Step 17	Test whether BCM is norm.
	1 Turn the ignition switch to OFF. 2 Disconnect and check the harness plug EB09_C、EB09_D、IP09_A、IP09_B of BCM. 3 Check and repair. • Corrosion(install new adapting piece or terminal-cleaning module and pin). • Pin damaged or bent-install new terminal/pin. • Fly out of pins-install new pins as needed. 4 Reconnect the BCM plug and all other previously disconnected plug. Make sure the pin is properly positioned and fixed. 5 Run the system and determine if the problem persists?
	YES → Replace the BCM.
NO ↓	
At this point the system is functioning normally. The problem may be caused by module connections. Address the root cause of any connector or pin problems,	
Next Step ↓	
Diagnosis end.	